Answers and marking guidance

Award one mark per correct, developed point unless stated. Other valid answers should be credited.

2 Describe two different ways the curved top could be created in 3D CAD.

- Extrude method: sketch the profile on a face or work plane, extrude, then fillet the top edge
- Revolve method: sketch the profile and revolve 90° about an axis
- Sizes or angle given

3 Quick check

- a) Centre axis and angle
- b) Loft
- c) Shell
- d) Revolve

4 Key terms

- 1 – B
- 2 – D
- 3 – A
- 4 – C
- 5 – E

5 Describe how the knob could be modelled using 3D CAD. You may use any suitable method.

- Revolve method: sketch the half profile with all sizes; identify the centre axis; revolve 360°
- OR loft method: Ø100 circle and Ø60 circle on a work plane offset 80; loft
- Sketch a Ø20 circle on the top face and extrude 12
- Sketch the slot with dimensions and extrude-subtract

6 Describe how the box would be modelled, in a sensible order.

- Sketch a 120 × 80 rectangle and extrude 60
- Fillet the four vertical edges R10
- Shell to 3 mm wall thickness
- removing the top face

No marks for: Shell applied before the fillet (the fillet would not follow inside) — order matters